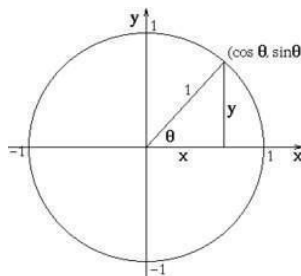
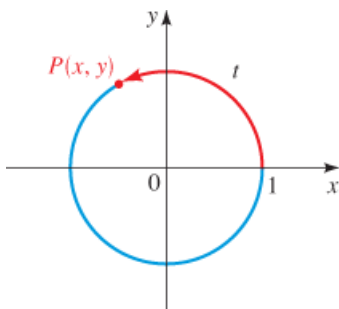


Chp 6.2 Trigonometry Functions of Real Numbers**I. The Trigonometric Functions**

The coordinates of any point on the unit circle are $(\cos t, \sin t)$

Definition of The Trigonometric Functions

Let t be any real number and let $P(x, y)$ be the terminal point on the unit circle determined by t . We define

$$\begin{aligned} \sin t &= y & \cos t &= x & \tan t &= \frac{y}{x} \quad (x \neq 0) \\ \csc t &= \frac{1}{y} \quad (y \neq 0) & \sec t &= \frac{1}{x} \quad (x \neq 0) & \cot t &= \frac{x}{y} \quad (y \neq 0) \end{aligned}$$

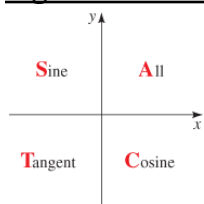
Because the trigonometric functions can be defined in terms of the unit circle, they are sometimes called the **circular functions**.

Domain of the Trigonometric Functions

Function	Domain
$\sin, \cos$	all real numbers.
$\tan, \sec$	all real numbers other than $\pi/2 + n\pi$ for any integer n
$\cot, \csc$	all real numbers other than $n\pi$ for any integer n

II. Values of the Trigonometric Functions

To compute other values of the trigonometric functions, we first determine **their signs**. The signs of the trigonometric functions depend on the quadrant in which the **terminal point** of t lies.

Signs of the Trigonometric Functions.

Quadrant	Positive Functions	Negative Functions
I	all	None
II	$\sin, \csc$	$\cos, \sec, \tan, \cot$
III	$\tan, \cot$	$\sin, \csc, \cos, \sec$
IV	$\cos, \sec$	$\sin, \csc, \tan, \cot$

Since the trigonometric functions are defined in terms of the coordinates of terminal points, we can use the **reference number** to find values of the trigonometric functions.

Evaluating Trigonometric Functions for Any Real Number

To find the values of the trigonometric functions for any real number, we carry out the following steps.

1. **Find the reference number.** Find the reference number associated with .
2. **Find the sign.** Determine the sign of the trigonometric function of by noting the quadrant in which the terminal point lies.
3. **Find the value.** The value of the trigonometric function of is the same, except possibly for sign, as the value of the trigonometric function of .

Reciprocal Relation

$$\csc t = \frac{1}{\sin t} \quad \sec t = \frac{1}{\cos t} \quad \cot t = \frac{1}{\tan t}$$

Even-Odd Properties

If $f(-x) = -f(x)$, odd

If $f(-x) = f(x)$, even

$$\begin{array}{lll} \sin(-t) = -\sin t & \cos(-t) = \cos t & \tan(-t) = -\tan t \\ \csc(-t) = -\csc t & \sec(-t) = \sec t & \cot(-t) = -\cot t \end{array}$$

The odd trigonometric functions are sine, cosecant, tangent, and cotangent. The even trigonometric functions are cosine and secant.

III. Fundamental Identities:

Reciprocal Identities

$$\csc t = \frac{1}{\sin t} \quad \sec t = \frac{1}{\cos t} \quad \cot t = \frac{1}{\tan t} \quad \tan t = \frac{\sin t}{\cos t} \quad \cot t = \frac{\cos t}{\sin t}$$

Pythagorean Identities

$$\sin^2 t + \cos^2 t = 1 \quad \tan^2 t + 1 = \sec^2 t \quad 1 + \cot^2 t = \csc^2 t$$

Example 1: Find the exact value of the trigonometric function at the given real number.

a. $\sin\left(-\frac{5\pi}{6}\right)$

b. $\tan(15\pi)$

c. $\csc \frac{7\pi}{3}$

d. $\sec \frac{11\pi}{6}$

Example 2: The terminal point determined by a real number t is given. Find $\sin t$, $\cos t$, and $\tan t$.

a. $\left(\frac{3}{5}, \frac{-4}{5}\right)$

b. $\left(\frac{-6}{7}, \frac{-\sqrt{13}}{7}\right)$

Example 3: Find the sign of the expression if the terminal point determined by t is in the given quadrant.

a. $\sin t \tan t$, Quadrant IV

b. $\frac{\tan t \sin t}{\cot t}$, Quadrant III

Example 4: From the information given, find the quadrant in which the terminal point determined by t lies.

a. $\csc t > 0$ and $\sec t < 0$

b. $\cos t > 0$ and $\cot t < 0$

Example 5: Write the first expression in terms of the second if the terminal point determined by t is in the given quadrant.

a. $\csc t$, $\cot t$; Quadrant II

b. $\tan^2 t$, $\sin t$; any quadrant

Example 6: Find the values of the trigonometric functions of t from the given information.

a. $\cos t = \frac{-7}{25}$, terminal point of t is in Quadrant III

b. $\tan t = -4$, $\csc t > 0$

c. $\sin t = \frac{-1}{4}$, $\sec t < 0$

Example 7: Find functions f and g such that $F = f \circ g$.

a. $F(x) = \cos^2 x$

b. $F(x) = \sqrt{1 + \tan x}$

Example 8: Determine whether the function is even, odd, or neither.

a. $f(x) = x^2 \cos(2x)$

b. $f(x) = x^3 + \cos x$

Example 9: The displacement from equilibrium ($y = 0$) of an oscillating mass attached to a spring is given by $y(t) = 4\cos(3\pi t)$ where y is measured in inches and t in seconds. Find the displacement at the times indicated in the table.

t	$y(t)$
0	
0.25	
0.50	
0.75	
1.00	
1.25	